

Problem Statement No.27- AI agent for chronic disease monitoring

The Challenge: An AI agent for chronic disease monitoring helps patients and healthcare providers manage long-term conditions effectively. It continuously analyzes health data from wearables, medical records, and patient inputs to detect early warning signs. Using AI and predictive analytics, it offers personalized insights, medication reminders, and lifestyle recommendations. The agent supports diseases like diabetes, hypertension, and heart conditions with real-time monitoring and alerts. It enables proactive care, reduces hospital visits, and improves patient adherence to treatment plans. This intelligent assistant bridges the gap between patients and providers, enhancing chronic care outcomes. Technology - Use of IBM cloud lite services /IBM Granity is mandatory.

The Objective: Build an AI-powered Chronic Disease Monitoring Agent that delivers real-time health monitoring, predictive alerts, and personalized care guidance for individuals, families, and health professionals. The solution should have: Real-Time Health Monitoring, RAG-Based Medical Knowledge, Predictive Risk, Alerts, Dual-Mode Reporting, Multi-Agent Architecture, Analysis Agent, RAG Knowledge Agent, Alert & Recommendation Agent, Longitudinal Dashboard.

Proposed Solution: CareWatch AI – Chronic Disease Monitoring Agent

The proposed solution is a multi-agent AI-powered Chronic Disease Monitoring System built using LangChain + CrewAI, IBM watsonx.ai, and IBM Granite Models to provide real-time chronic disease monitoring, predictive alerts, and personalized care guidance. The system addresses the critical gap in continuous patient monitoring through an intelligent, modular multi-agent architecture.